MIL-D-24542A
31 January 1980
SUPERSEDING
MIL-D-24542(SH)
5 October 1976
(See 6.7)

MILITARY SPECIFICATION

DIVING OUTFIT, LIGHTWEIGHT

* This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a lightweight diving outfit utilizing a demand breathing device for use in diving operations, employing two divers and a standby diver, to 190 feet when air is used for breathing and, when additionally supported by an open diving bell, to 300 feet when helium-oxygen gas mixture is used for breathing.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Edged and Components.
PPP-B-676 - Boxes, Set-up.
PPP-C-1842 - Cushioning Material, Plastic, Open Cell (For Packaging Application).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD For 1426) appearing at the end of this document or by letter.

FSC 4220

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FEDERAL (Continued)

PPP-P-40 Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-117 - Bags, Sleeves and Tubing-Interior Packaging.
MIL-R-196 - Repair Parts, Accessories, and Kits, Mechanical Packaging of.
MIL-B-233 - Boxes, Repair Parts, Storage.
MIL-P-775 - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric or Metal (including Tubing), and Fittings, Nozzles and Strainers.
MIL-C-3131 - Cordage; Preparation for Delivery of.
MIL-C-12000 - Cable, Cord and Wire, Electric, Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment Accessories, and Repair Parts, Packaging and Packing of.
MIL-B-81705 - Barrier Materials, Flexible, Electrostatic-Free, Heat Sealable.
MIL-P-81997 - Pouches, Cushioned, Flexible, Electrostatic Free, Reclosable, Transparent.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MS90363 - Box, Fiberboard, with Cushioning for Special, Minimum Cube Storage and Limited Reuse Application.

DATA LISTS (DL)

NAVAL SEA SYSTEMS COMMAND

NAVSHIPS DL 4666881 - Divers Mask USN MK1 MOD 0.
NAVSHIPS DL 4666883 - Umbilical, Divers Mask USN MK1 MOD 0.
NAVSHIPS DL 4666884 - Setup/Maintenance Kit, Divers Mask USN MK1 MOD 0.
NAVSHIPS DL 4666887 - Operating Specifications, Divers Mask USN MK1 MOD 0.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND

NAVSHIPS 4666885 - Divers Intercom.

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PUBLICATION[1/]
NAVAL SEA SYSTEMS COMMAND
OD 46363 - Requirements for the Electrostatic Discharge Protection of Electronic Components and Assemblies.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

\# 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification, Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee, Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA Traffic Dept., 1616 P Street, N.W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.1.1), the contractor shall furnish sample units(s) for first article inspection and approval (see 4.3 and 6.2).

\# 3.2 Construction. The lightweight diving outfit shall consist of equipment necessary to support a diving operation employing two working divers and one standby diver. The outfit shall include four diver's mask assemblies (see 3.3), three

1/ Copies may be obtained from Code 032, Naval Ocean Systems Center, San Diego, CA 92152.

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diver's umbilical assemblies (see 3.4), a diver's intercommunications set (see 3.5), a set-up and maintenance kit (see 3.6), and operating space items (see 3.7). A reusable container shall be furnished that provides stowage for certain parts of the outfit (see 5.1.2.1).

3.3 Diver's mask assembly. The diver's mask assembly shall be in accordance with the drawings listed on DL 4666881.

3.3.1 Special tooling. If specified (see 6.1.1), molds, jigs, and dies for production of the mask frame and of the oral-nasal mask will be made available to the contractor "as is, and where is". This special tooling will be available for viewing (as is, where is) by prospective bidders on a contract for the purpose of preparation of their bids. At completion of a contract under which the contractor has taken delivery of the special tooling, the special tooling shall be redelivered to the Government in condition "as good as received, less normal wear and tear".

3.3.2 Mask control. The mask control shall be capable of withstanding the mask control performance test of 4.6.2.

3.4 Diver's umbilical assembly. The diver's umbilical assembly shall be in accordance with the drawings listed on DL 4666883. The assembly shall be capable of withstanding the continuity test of 4.6.1. When specified (see 6.1.1), this item shall be delivered disassembled.

3.5 Diver's intercommunications set. The diver's intercommunications set will be an item furnished by the Government (GFE, see 3.9). The essential mechanical and dimensional characteristics of this item are shown on Drawing 4666885.

* 3.6 Set-up and maintenance kit. The set-up (including tools) and maintenance kit shall be in accordance with the drawing listed on DL 4666884.

* 3.7 Operating space items. Operating space items shall include those items listed on DL 4666887, in the quantity indicated thereon, and also the regulator operating space item kits (see 3.7.1 and 3.7.2) and the cylinder valve operating space item kit (see 3.7.3).

* 3.7.1 Demand regulator operating space item kit. The demand regulator operating space item kit shall contain twelve replacements for each part of the demand regulator that is fabricated, entirely or in part, from plastic or rubber, with the exception of the exhaust tube. The kit shall contain two replacement exhaust tubes, four replacement springs for each spring used in the demand regulator, and four replacement actuating levers ("horse-shoes"). The kit shall also contain six replacement inlet fittings (low pressure nozzles or inlet nipples) and six replacements for each nut and each spring-type

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retainer used in the demand regulator. The individual parts, unit packaged in accordance with the requirements of 5.5, shall be additionally marked with their identifying numbers as used on the demand regulator maintenance instruction sheet (see 3.10), and consolidated with the maintenance instruction sheet into kit form. This kit shall be marked with a manufacturer's part number.

* 3.7.2 First stage regulator operating space item kit. The first stage regulator operating space item kit shall contain twelve replacement parts for each replaceable part of the first stage regulator that is fabricated entirely or in part, from plastic or rubber, with the exception of the flexible hose assembly. The kit shall contain four replacement flexible hose assemblies (that is, hose with end fittings), four replacement high pressure nozzles (or that smallest or most basic subassembly that contains the high pressure nozzle), four replacement spring-type retainers, four replacement regulator inlet (sintered) filters, and four replacement relief valves (or that smallest or most basic subassembly that contains the relief valve). The individual parts, unit packaged in accordance with the requirements of 5.5, shall be additionally marked with their identifying numbers as used on the first stage regulator maintenance instruction sheet (see 3.10), and consolidated with the maintenance instruction sheet into kit form. This kit shall be marked with a manufacturer's part number.

* 3.7.3 Cylinder valve operating space item kit. The cylinder valve operating space item kit shall contain twelve replacement parts for each replaceable part of the valve assembly that is fabricated, either entirely or in part, from plastic or rubber and twelve replacements for each washer. There shall also be four replacement valve stems and retainer nuts (or other retainer that holds the valve handle to the valve stem) and four of each piece required to completely replace the safety valve (or safety plug). The individual parts, unit packaged in accordance with the requirements of 5.5, shall be additionally marked with their identifying numbers as used on the cylinder valve maintenance instruction sheet (see 3.10), and consolidated with the maintenance instruction sheet into kit form. This kit shall be marked with a manufacturer's part number.

3.8 Controlled source part. On all equipment delivered under one contract, a part that is described by source control documentation shall be of a single source and model.

3.9 Government-furnished. The following items are Government-furnished properties for incorporation into the lightweight diving outfit furnished under this specification.

(a) Diver's intercommunications set (see 3.5).
(b) Diver's intercommunications set operations and technical manual (see 3.11).

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(c) Lightweight diving outfit technical manual (see 3.11).

The contractor shall notify the contracting officer of his requirement for each of items (a), (b), and (c) and of the last acceptable date for delivery of these items to his plant for incorporation into the lightweight diving outfit. Such notification shall be given in time to allow 180 days for delivery of the items to the contractor's plant by the Government prior to that last acceptable date.

* 3.10 Regulators and cylinder valve maintenance instructions. A copy of each of the manufacturer's commercial maintenance instructions, including illustrated parts breakdown (exploded view), for the demand regulator, for the first stage regulator and for the cylinder valve used in the outfit shall be furnished packaged with the operating space item kits for these items (see 3.7.1, 3.7.2 and 3.7.3).

* 3.11 Technical manuals. Two technical manuals shall be packaged with the diving outfit, one covering the lightweight diving outfit in general, and the other covering the diver's intercommunications set. These manuals will be items furnished to the contractor by the Government (GFP, see 3.9). They shall be packaged with the outfit as required herein (see 5.7).

3.12 Workmanship. Equipment shall be free of defects such as cracks or flaws in material. Made-up joints shall be leak-tight. Air (gas) channels shall be clear and free of collars (or steps), burrs, and other protrusions into flow paths. There shall be no gross tool marks. All mold flashings shall be removed. Polished external surfaces shall have a consistency and evenness of finish. Assembled equipment shall be free of chips, filings and other residues resulting from manufacturing processes. Internal components, parts, and pipes shall be cleaned of grease, oil, and other material deleterious to its use with high pressure air. Except where otherwise specified, colors shall be even. The equipment shall be free of burrs, sharp corners, and edges that will injure human skin or puncture pure gum sheet rubber.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection system plan. The contractor shall prepare an inspection system plan in accordance with the data ordering document included in the contract (see 6.1.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

* 4.3 First article inspection. First article inspection shall be performed on one complete lightweight diving outfit and shall consist of the examination and tests specified in table I.

TABLE I. First article inspection.

Examination and test	Requirement paragraph	Test paragraph
Visual examination	3.12	4.5
Umbilical continuity	3.4	4.6.1
Mask control performance	3.3.2	4.6.2
Preparation for delivery inspection	-----	4.7

4.3.1 First article inspection report. The contractor shall prepare a first article inspection report in accordance with the data ordering document included in the contract (see 6.1.2).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the production inspection of 4.4.1.

4.4.1 Production inspection. Each diving outfit shall be visually examined in accordance with 4.5 and each mask shall be subjected to the mask control performance test of 4.6.2.

4.5 Visual examination. The sample shall be visually examined to assure conformance with the requirements of 3.12.

4.6 Tests.

4.6.1 Umbilical continuity. Electrical continuity between corresponding pins on opposite ends of the communications cable shall be determined.

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4.6.2 Mask control performance. The controls of each sample mask (see figure 1) shall be subjected to the following test, using dry water-pump air or nitrogen in the sequence specified:

(a) Place the mask on a test fixture that will isolate the oral nasal cavity from the mask frame cavity and both cavities from ambient pressure. Attach a compressed air supply with a capacity of at least 40 standard cubic feet per minute (std ft^3/min) in the gage pressure range of 60 to 180 lb/in^2, being adjustable for pressure within that range, to the breathing air inlet of the mask (see figure 1).

(b) With valves A and B closed, set the air supply to a gage pressure of 60 + 0, -10 lb/in^2 and the demand regulator adjustment to the "full-out" position. The regulator shall free flow.

(c) Reset the demand regulator adjustment to the "full-in" position and the air supply to a gage pressure of 180 + 10, -0 lb/in^2. The regulator shall not free flow.

(d) Reset the air supply to a gage pressure of 125 $\pm$ 5 lb/in^2 and open valve B all the way. Flow into the mask through valve B shall be not less than 35 ft^3/min.

(e) Close valve B. Reset the demand regulator adjustment so that the regulator starts to deliver air when a partial vacuum of 1.50 $\pm$ 0.10 inches of water is applied to the oral nasal cavity. Increase the partial vacuum in the oral nasal cavity to 3.00 $\pm$ 0.10 inches of water. The flow into the mask through the regulator shall be measured to be not less than 15 ft^3/min.

(f) Reset the air supply to a gage pressure of 180 + 10, -0 lb/in^2. Adjust the demand regulator adjustment so that the regulator does not free flow. Immerse the mask and test fixture in water to cover. Depress the regulator purge button until there is a steady air flow from both the mask frame exhaust valve and the regulator exhaust valve and the regulator exhaust tube. The mask shall be free of air leaks (air leaks that can be attributed to the test fixture or to its seals with the mask are of no consequence).

4.7 Inspection of preparation for delivery. Packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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* 4.7.1 First article sample. The contractor shall conduct inspection on one complete package, packed for shipment, to ascertain that the cleaning, drying, preservation, packaging, packing, and marking of the equipment or item conforms to this specification and the quality conformance inspection requirements of MIL-B-233 (see 6.2). First article inspection shall be repeated when changes in preservation and packing materials, or processes, or designs are made.

4.7.2 Dummy load. When specified (see 6.1.1), a dummy or simulated load, packaged and packed for shipment may be used for the first article inspection (see 4.7.1).

4.8 Government examination of last five items. In the event that Government furnished special tooling (see 3.3.1 and 6.1.1) has been used in production of equipment, the last five mask frames and oral-nasal masks produced will be examined by the Government inspector to determine that the condition of Government furnished special tooling is in accordance with the requirements of 3.3.1.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 General requirements.

5.1.1 Technical data.

* 5.1.1.1 Drawing details and test report. When specified (see 6.1.1), complete description details on drawings, preservation, packaging, packing and marking of the first article pack shall be prepared in accordance with the data ordering documents (see 6.1.2) and submitted for approval to the command or agency concerned, via the packaging element of the administrative contract office (ACO). Approval will not relieve the contractor of inspection specified in 4.7.1. Upon completion of the first article testing, a detailed test report shall be forwarded to the command or agency concerned (see 6.1.2).

* 5.1.1.2 Packaging and transportation data. When specified (see 6.1.1), packaging and transportation data in accordance with the data ordering document (see 6.1.2), shall be submitted.

5.1.2 Container design criteria. Lightweight diving outfits shall be prepared for delivery in a three-container concept as specified herein.

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* 5.1.2.1 Container number one. The container shall conform to type M of MIL-B-233, except for size and the following:

(a) The container size shall be the minimum necessary to pack the following equipment of the outfit:
(1) Four diver mask assemblies without backplate and cylinders consisting of mask with hood, spider and insert (see table II, Note 2).
(2) One diver's intercommunications set.
(3) One set-up, maintenance kit.
(4) One kit of miscellaneous parts for the umbilical assembly (such as D-rings, tape, and tie cord).
(5) Operating space items (see 3.7 and 5.5).
(b) The container shall be provided with accessory framing trays, partitions, compartments, or other means that will provide space for parts storage with ease of removal or replacement.
(c) The container shall be provided with ventilation, splash protection, and drainage with all openings covered with a corrosion-resistant wire cloth or screening that will prevent entrance of vermin or rodents.
(d) Nominal 2- by 4-inch wood skids shall be bolted to the base of the container.
(e) Rope handles shall not be used.

* 5.1.2.2 Container number two. The container shall conform to 5.3 for the level specified, and shall contain four cylinders, with filler valves and four cylinder harnesses. Cylinders shall be assembled to the harness.

* 5.1.2.3 Container number three. The container shall conform to 5.3 for the level specified and shall contain three complete divers umbilicals, assembled or disassembled as required (see 3.4 and 6.1.1).

5.1.3 Closure of openings. All openings into valves, valve breathing mechanisms, or hose ends (with or without fittings), shall be sealed with plastic caps or by wrapping the opening with a flexible waterproof barrier. The flexible waterproof barrier shall be secured with waterproof pressure sensitive tape to prevent entrance of dust, dirt, or any foreign material.

5.1.4 Cushioning, anchoring, blocking, and bracing. Cushioning, anchoring, blocking, and bracing of the contents of all containers shall be provided to insure that the contents are protected against damage from abrasion or deformation during handling, shipment, and storage. Guidelines for cushioning, anchoring, blocking, and bracing are specified in MIL-STD-1186. Excelsior, shredded newspaper, and similar materials which have

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not been chemically refined for non-corrosiveness are unacceptable unless they are enclosed in a sealed waterproof barrier, bag, or envelope. "Loose fill" material is prohibited for packaging and packing applications such as cushioning, filler, or as dunnage material.

* 5.1.5 Use of dusting material. Talc/talcum and soapstone powders used in the packaging process of item(s) shall be free of asbestos.

5.2 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. All equipment of the diving outfit shall be thoroughly cleaned and dried in accordance with the applicable requirements of MIL-P-116 which will insure that all parts are clean, dry, and free of grease, oil, and any other foreign matter that might adversely affect the operation of the outfit.

5.2.1.2 Unit protection. Unless otherwise specified (see 6.1.1), assemblies of each lightweight diver's outfit shall be individually unit protected in accordance with the requirements of table II.

TABLE II. Preservation-packaging (methods or sub-methods).

Item subassembly	Method or sub-method of MIL-P-116[1/]	Other packaging process specification
Divers mask	IC-3[2/]	-----
First stage regulator and hose	IC-3	-----
Safety harness	III	-----
Set-up/maintenance kit	-----	PPP-P-40
Intercom box	IC-3	-----
Cylinder w/filler valve	III[3/]	-----
Cylinder harness	III	-----
Divers umbilical		
-Assembled		
-Disassembled:	III	-----
Marrying tape	IC-1	-----
Telecord splice kit	IC-3	-----
Telecord	-----	MIL-C-12000
Life line w/finished ends, "D" rings and belt hook	-----	MIL-C-3131
Hose, air or pneufathometer w/fittings attached	-----	MIL-P-775
Marrying cord	III	-----

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1/ Bags, sleeves, or envelopes utilized for methods IC-1, IC-3, and method III shall be of transparent material as specified in MIL-P-116.
2/ Each diver's mask and hood assembly, prior to packaging (see 5.2.1.2 and 5.2.2), shall be furnished with an insert to provide support to prevent creasing or crushing of the hood material, and to serve as a handling, drying and storage fixture.
3/ Cylinders shall be pressurized a minimum gage pressure of 3 pounds per square inch (lb/in^2) with dry air.

5.2.1.3 Unit and intermediate containers. Like items, unit-protected as specified in table II, may be individually packaged in unit or combined in intermediate containers. Unit and intermediate containers shall be of folding, set-up paperboard, fiberboard, or metal-stayed paperboard conforming to PPP-B-566, PPP-B-636, PPP-B-665, and PPP-B-676, respectively, with selection at the option of the contractor. The gross weight of paperboard boxes shall not exceed 10 pounds. Unless otherwise specified (see 6.1.1) the gross weight of fiberboard boxes shall not exceed 20 pounds. Box closure shall be as specified in the applicable box specification or appendix thereto except that when class weather-resistant boxes conforming to PPP-B-636 are used, method V closure in the appendix to PPP-B-636 shall apply.

5.2.2 Level C. Packaging of lightweight diving outfits shall afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal preservation-packaging methods may be utilized when such meet the requirements of this level. In addition, the diver's mask and hood shall, as a minimum, be provided with an insert as specified under table II (Note 2).

5.2.3 Government-furnished special tooling. If the contractor has taken delivery of Government-furnished special tooling (see 3.3.1), at the completion of the contract, he shall reclean the special tooling, represerve it in its original manner and return it to its packaging, packing, and marking. The packaging, packing and marking shall be restored to the "as new" condition.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.1.1).

5.3.1 Levels A and B. Exterior (shipping and storage) containers shall conform to table III. Unless otherwise specified (see 6.1.1) the container selection under the level of packing specified, shall be at the contractor's option. Container number one (see 5.1.2.1) shall not require over-packing.

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TABLE III. Container selection for packing.

Specification	Container	Packing application	
		Level A	Level B
		(Style, type, or class)	
PPP-B-601	Wood, cleated plywood	Overseas type	Domestic type
PPP-B-621	Wood, nailed and lockcorner	Class 2 overseas	Class 1 domestic

NOTES:
1. Maximum gross weight, container plus contents, shall not exceed the applicable requirements for the style, type, or class container selected for the level of packing to be applied.
2. Shipping containers shall be closed, strapped, or banded in accordance with the applicable container specification, or appendix thereto.

5.3.2 Level C. Lightweight diving outfits packaged as specified, shall be packed in containers acceptable to the common carrier which will insure safe delivery at the destination in a satisfactory condition at the lowest applicable rate. Containers or method of packing shall comply with Uniform Freight or National Motor Freight classification rules or regulations or other carrier rules as applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required (see 6.1.1) marking shall be in accordance with MIL-STD-129 and MIL-P-116.

5.4.1 Special marking. In addition to the marking required by MIL-B-233, container number one (see 5.1.2.1) shall be marked "REUSABLE CONTAINER-DO NOT DESTROY".

* 5.5 Operating space items and tools. Operating space items and tools shall be preserved-packaged, packed, and marked in accordance with the levels of PPP-P-40, MIL-C-3135, MIL-R-196, or MIL-E-17555, as applicable, for the intended use and destination as shown in table IV.

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TABLE IV. Preservation-packaging and packing of operating space items and tools.

Intended use-destination	Levels	
	Preservation-packaging	Packing
Accompanying equipment	A	1/
Stock	A	B
Immediate use	C	C

1/ Unless otherwise specified (see 6.1.1), operating space items shall be packed in container number one (see 5.1.2.1).

* 5.5.1 Repairables. Repairable operating space items subject for return to a repair facility for restoration and reissue shall be packaged and packed in materials and containers capable of reuse. Unless otherwise specified (see 6.1.1), packaging shall be in accordance with MS 90363. Reusable containers, when required repairables, shall be in accordance with the guidelines of MIL-E-17555. Containers shall be marked "REUSABLE DO NOT DESTROY". Unless otherwise specified (see 6.1.1), packaging and transportation data in accordance with the data ordering document (see 6.1.2) shall be submitted for items falling in this category.

* 5.6 Sensitive electronic item protection.

* 5.6.1 Sensitive electronic item protection. Items such as, but not limited to, diodes, transistors, integrated circuits, and equipments incorporating such items which are susceptible to damage from electrostatic, electromagnetic or both field forces shall be protected against such damage as follows:

* 5.6.2 General requirements.

(a) Lead and terminal protection. Leads or terminals configuration shall be maintained as manufactured without causing loads or stresses capable of causing damage to the item. Protection shall be by means of a carrier, container design, or inserts of noncorrosive electrostatic free supporting materials. Materials used to maintain item position and lead or terminal configuration shall permit item removal and replacement without damage to the item.

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(1) Carrier. Carriers, when used for additional protection of miniature electronic items, shall be of such strength to prevent damaging resonances, shocks, and electrostatic charges to the sensitive item(s). Anchoring or securing of the item, leads or terminal within the carrier by means of tape or adhesive is prohibited. The carrier shall maintain physical separation and manufactured configuration of the item leads or terminals during packaging, handling, transportation, storage and stowage and for testing operations. The carrier shall be designed to permit safe and easy removal, inspection, and item replacement, and designed without sharp edges to preclude subsequent damage to the item and packaging material or method.

(b) Wraps and cushioning. Wraps and cushioning, when required for additonal protection, shall be noncorrosive and in compliance with the requirements of MIL-P-116, and shall not crumble, flake, powder or shed. Wraps or cushioning direct contact with electrostatic sensitive item(s) shall conform to the electrostatic protection requirements specified herein.

* 5.6.3 Specific requirements.

* 5.6.3.1 Electrostatic protection. Item(s) adversely affected by electrostatic field forces shall be provided an initial wrap of material conforming to MIL-B-81705, Type II, or cushioned in material conforming to PPP-C-1842, Type III, Style A or B, and unit packaged in heat-sealed bags conforming to MIL-B-117, Type I, Class F, Style 1. Alternatively, reclosable cushioned pouches conforming to MIL-P-81997, Type I or II may be used in lieu of the initial wrap or cushioning. Noncorrosive conductive material(s) shall be applied to all exposed leads and connector pins to maintain a common potential. This is to protect the item(s) from electrostatic charge(s) that may be encountered during handling. Packaging materials currently covered by title, scope, or intended use under Government specification(s), but modified as electrostatic-free material(s), or newly developed electrostatic-free packaging material(s) not covered by a Government packaging material specification(s) are encouraged for use. Use of such modified or newly developed electrostatic-free packaging material(s) will be permitted subject to the contracting officer's determination that (a) the physical properties of such material(s) are equal to or better than similar constructed material(s) covered under a required Government packaging material specification and, (b) that such materials satisfy the electrostatic decay rate requirement of MIL-B-81705. The material manufacturer or contractor

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shall furnish to the Government Inspector for review documented proof of conformance to the requirements specified herein, certified by an acceptable testing laboratory. Upon submission for acceptance, copies of the contractor's documented proof shall be forwarded to the procurement contracting officer (PCO) packaging activity. The decision of the contracting officer shall be final as the acceptability or non-acceptability of the packaging material and that decision shall not be subject to review under the disputes clause of the contract. When such materials are acceptable, unit packaging shall be in accordance with the procedures for electrostatic protection.

5.6.3.2 Electromagnetic protection. Unless otherwise specified in the contract, item(s) subject to damage by electromagnetic forces shall be unit packaged in heat-sealed, barrier bags conforming to MIL-B-117, Type I or II, Class E, Style 1 or Type I, Class F, Style 1. When MIL-B-117, Type 1 or II, Class E, Style 1 bags are selected and used, the barrier material shall also contain a laminate of aluminum foil as well as meeting the requirements of MIL-B-117.

5.6.3.3 Electromagnetic and electrostatic protection. When the item(s) requires both electromagnetic and electrostatic protection, unit packaging shall be as specified under "Electrostatic Protection".

5.6.3.4 Marking. In addition, to the marking required by MIL-STD-129 (see 5.4) and the contract, the MIL-STD-129 markings for sensitive electronic items shall apply except as follows: The sensitive electronic device caution label, 2 inch by 2 inch or less, specified for intermediate packages shall be used for unit packages in lieu of the device symbol. In addition, special caution markings shall be placed on all unit and intermediate (interior) packages and shipping containers. Electrostatic sensitive item caution markings may be placed adjacent to the MIL-STD-129 sensitive electronic device caution label. Caution marking shall be as follows:

"CAUTION--ELECTROSTATIC SENSITIVE DEVICE:
Remove electrostatic protection at use or in protected area. Reuse packaging materials for the unservicable item. See NAVSEA Publication OD 46363 for protective handling for this item."

5.7 Technical manuals. Technical manuals which accompany shipments that are packed level A or B shall be packaged in a transparent waterproof plastic bag, minimum 4 mils thick. Closure shall be by heat sealing. Technical manuals shall not be placed within any flexible sealed barrier enclosing parts. The copy(s) of the manual shall be placed in container number

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one (see 5.1.2.1). Packing lists shall indicate which container contains the technical manuals, and shall also state the approximate location therein. For ease of removability, the location of the manual shall be such that it is readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specification or packed in containers conforming to the requirements for level A, B, or C, as specified.

5.8 Data. Drawings, microfilm, miscellaneous reports, to be forwarded to the Naval Sea Systems Command in compliance with the requirements of this specification or as specified in the contract data requirements list, and such, (see 6.1.2), shall be packaged and packed level C in accordance with the applicable data specification, or where no preparation for delivery requirements exist, data shall be prepared in accordance with the level C requirements specified herein.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is required for inspection and approval (see 3.1, 4.3, and 6.2).
(c) If Government-furnished equipment will be furnished (see 3.3.1).
(d) When umbilical is to be delivered disassembled (see 3.4 and 5.1.2.3).
(e) When a dummy or simulated load may be used for first article inspection (see 4.7.2).
(f) Government examination required for last five items (see 4.8).
(g) When complete description details on drawings, preservation, packaging, packing and marking of the first article pack are required (see 5.1.1.1).
(h) When packaging and transportation data is required (see 5.1.1.2 and 5.5.1).
(i) Levels of preservation-packaging and packing required (see 5.2 and 5.3).
(j) Unit protection other than the requirements of table II (see 5.2.1.2).
(k) Gross weight, other than specified, for fiberboard boxes (see 5.2.1.3).
(l) Container selection other than contractor's option (see 5.3.1).

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(m) Special marking, if required (see 5.4).
* (n) Packing of operating space items other than specified (see 5.5).
* (o) Packaging of operating space items containers, if other than as specified (see 5.5.1).

* 6.1.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
4.1.1	Inspection system plan	UDI-R-23574	
4.3.1	First article inspection report	UDI-T-23450	
5.1.1.1	Drawings, engineering and associated list	DI-E-7031	Level 2 for Preservation-Packaging, Packing and Marking of Standard packages Level 3 for reusable containers
5.1.1.1	Report, Preservation-Packaging and Packing Test	UDI-T-23766	
5.1.1.2	Packaging and Transportation Support Data	UDI-P-23508	

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

* 6.1.2.1 The data requirements of 6.1.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

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6.2 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a lightweight diving outfit and packaging, which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Special tooling. The contracting activity should assure that special tooling (see 3.3.1) is available for viewing by prospective bidders on the contract for preparation of their bids. It should also assure availability of the special tooling to the successful contractor in the time frame required by the contract. Finally, the contracting officer should assure that the special tooling is returned to the Government at completion of the contract, or when the successful bidder has declined its use (see 5.2.3).

6.5 Government-furnished material. The contracting officer should arrange to furnish the items listed in 3.9.

* 6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

* 6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment.

6.7 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - ME
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Preparing activity:
Navy - SH
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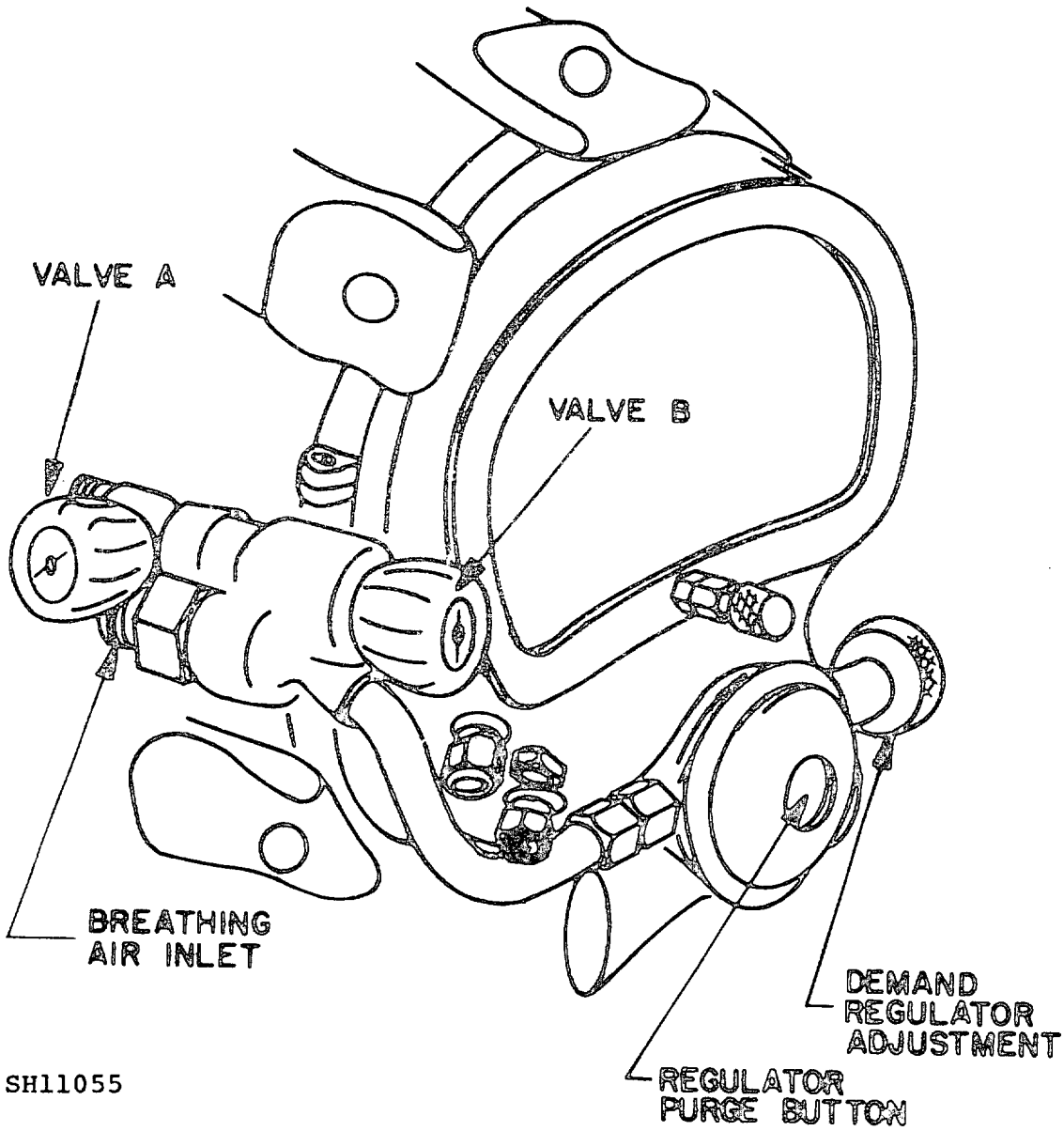


FIGURE 1. Mask control arrangement.

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